

A Packing of 18 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 18 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.352614. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.345324, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 18 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.352614 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 18$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.345324, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 18/17 = 2.352941$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

```
python grassmann/verify.py grassmann/packings/m9n4N18.txt
```

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
147038430066747604664595780220022462557393060349239493965411663851547206
716106607870884664678236755073140785621511991387019563635719177460350802
126271534297741581275330698204156799128463107365
```

denominator:

```
625000000000092815308791615902933129657730390168068763154003545024337522
510474081669901271437050728414626221806734134124233688629006502628677964
00570128094364042883870510088558895881002250096
```

B Coordinates

18 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N18.txt`.

```
-188124875316 -75569180849 -136492199676 -206407342166204913900 -208199213434 -110163533545 24405040671436975471382 -81523550955 -333049982085 -262754749286
-93052422345 -362530450239 -504962720619 -433490551363931551108 239006242913 767183183962 41468210909207145374987 221392593059 -415122989270 421170231049
133583125033 321380596116 -161776838815 5161359232273135563039 -444396709459 299079758735 163597593174726732632474 70691403653 57942059901 -532602093658
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-127027434170 286533130070 45726231955 -45633271755948849095581 -85420549609 339917924110 -5060964802021171354800 452971925191 194592426080 258600217164
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```